CUSTOMER CHURN & RETENTION · COSMETICS E-COMMERCE

85% of buyers never came back. So who's worth saving?

20.7M clickstream events · 110K buyers

REES46 store, Oct 2019 – Feb 2020

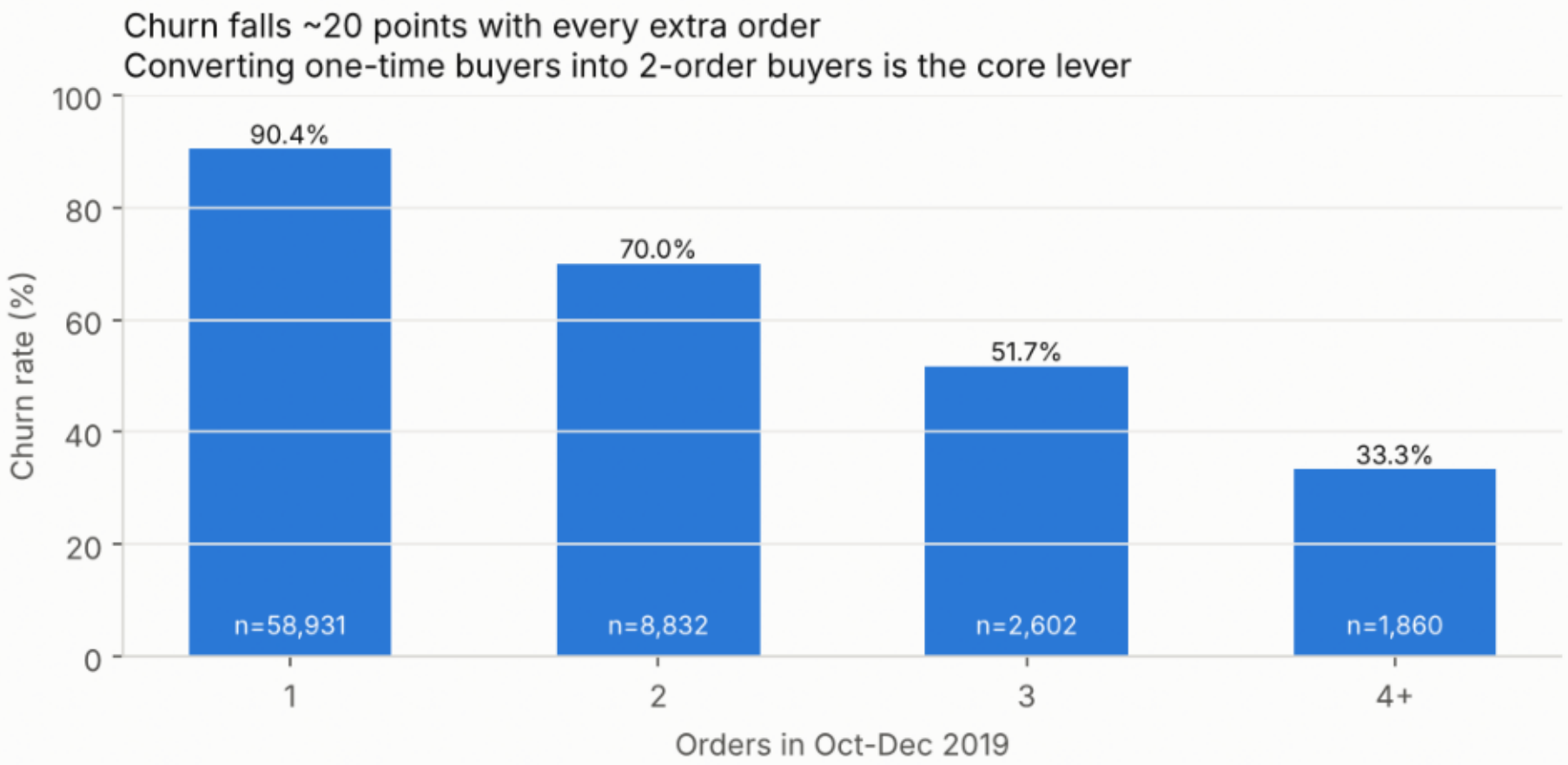
SQL · Python · XGBoost · Power BI · Excel

Buyers who didn't return brought 74% of Oct–Dec revenue

85% of Oct–Dec buyers placed no order in Jan–Feb.

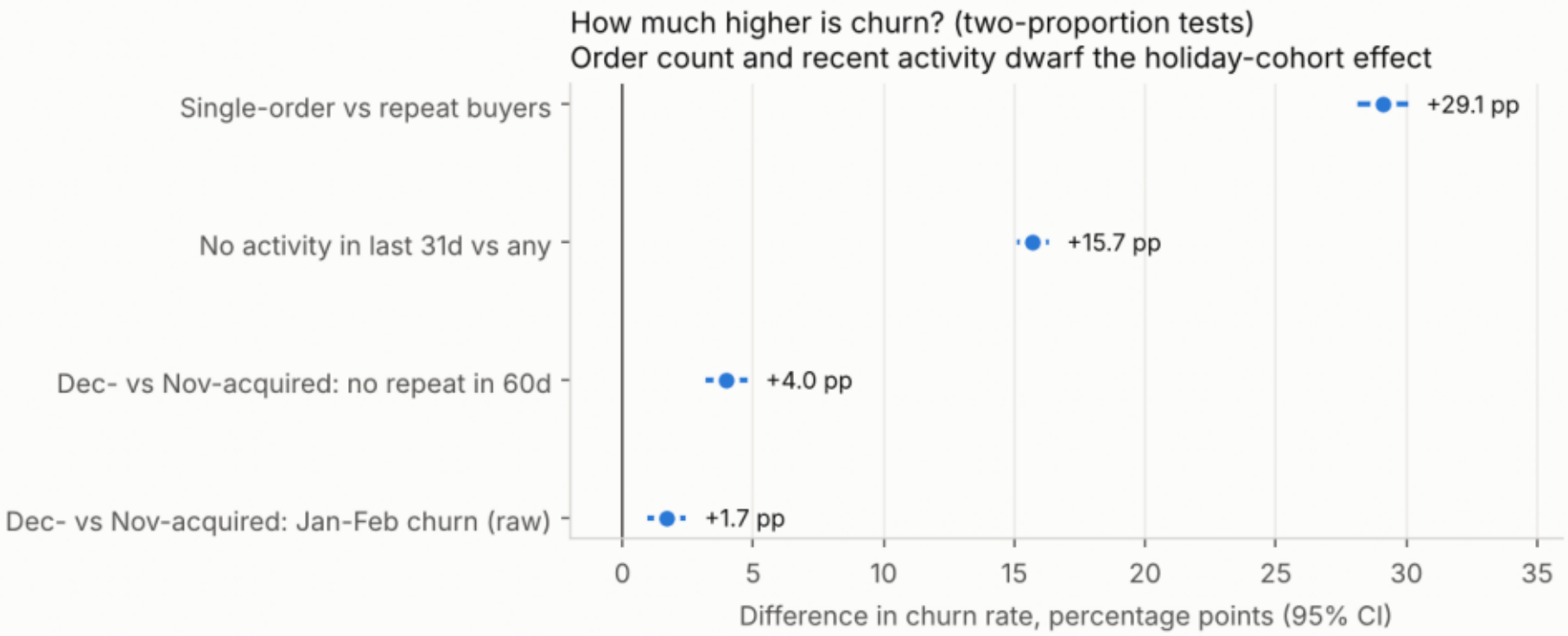
One-time buyers churned at 90%, repeat buyers at 61%

Every extra order lowers churn, so the second order is the target.



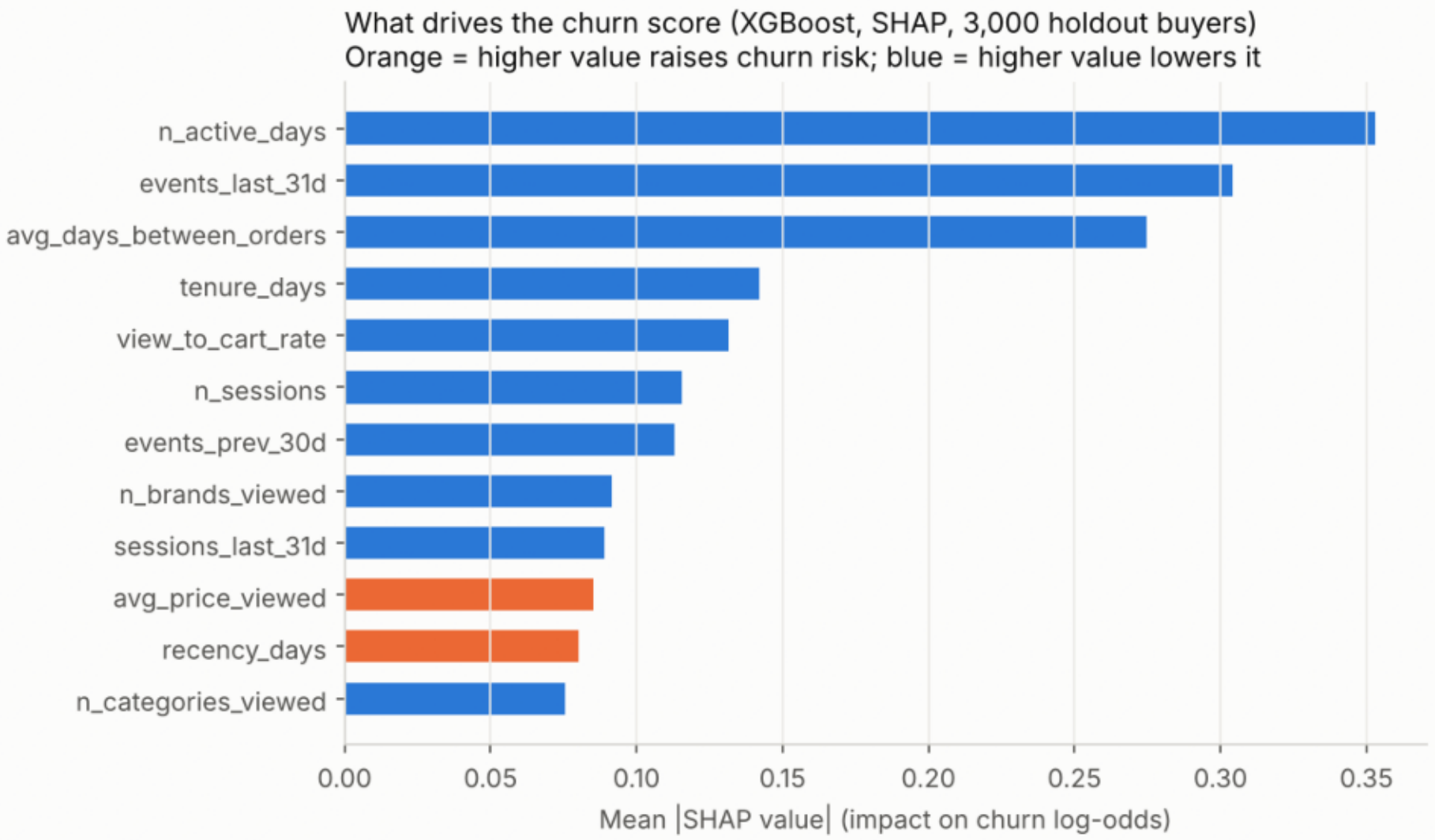
A silent last month adds about 16 points of churn

Cart removals did not predict churn: they signal engaged shoppers.



XGBoost ranks risk with AUC 0.80 (vs 0.73 for an RFM baseline)

"Everyone churns" is 85% accurate and useless, so I used AUC, PR-AUC and lift.

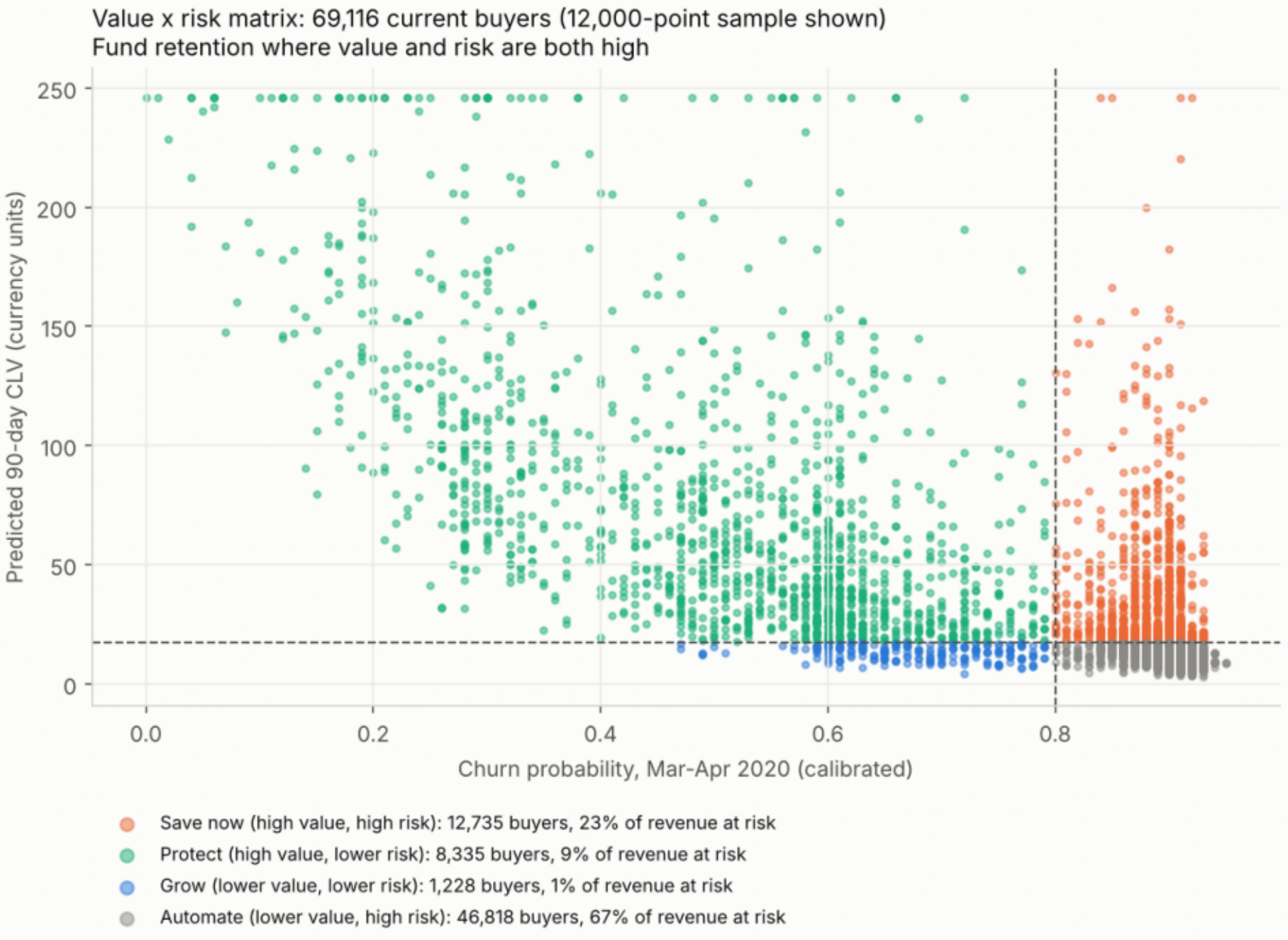


5 · WHO TO SAVE

Target 12.7K high-value, high-risk buyers: projected +81% ROI

About 14.6K net per wave; breaks even below a 5% save rate.

Blanket discounts only break even.



6 · WHAT I'D DO NEXT

Three quick wins now, one big bet tested properly

Win-back, no-discount nudges and cart recovery, each with a holdout.
A 10% offer to every new buyer needs a ~10-point lift,
so it goes to a 3-arm A/B test first.

